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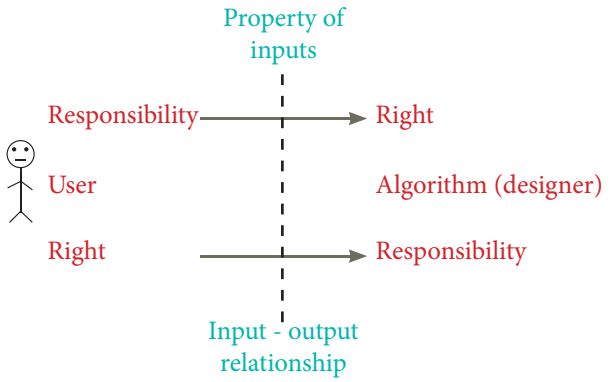

Figure 6.2: Input property and the input-output relation as rights and responsibilities

Example 6.8. Consider the specification of the algorithm square_root.

square_root(n)

-- inputs: n is a real number, n ≥ 0.

-- **outputs : y is a real number such that $y^2 = n$.**

The algorithm designer can assume that the given number is non-negative, and construct the algorithm. The user can expect the output to be the square root of the given number.

The output could be the negative square root of the given number. The specification did not commit that the output is the positive square root. If the user passes a negative number as the input, then the output need not be the square root of the number.

6.6 Abstraction

To ride a bicycle, it is sufficient to understand the functioning of the pedal, handlebar, brakes and bell. As a rider, we model a bicycle by these four features. A bicycle has a lot more details, which the rider can ignore. Those details are irrelevant for the purpose of riding a bicycles.

A problem can involve a lot of details. Several of these details are irrelevant for solving the problem. Only a few details are essential. Abstraction is the process of ignoring or hiding irrelevant details and modeling a problem only by its essential features. In our everyday life, we use abstractions unconsciously to handle complexity. Abstraction is the most effective mental tool used for managing complexity. If we do not abstract a problem adequately, we may deal with unnecessary details and complicate the solution.

Example 6.9. A map is an abstraction of the things we find on the ground. We do not represent every detail on the ground. The map-maker picks out the details that we need to know. Different maps are drawn for different purposes and so use different abstractions, i.e., they hide or represent different features. A road map is designed for drivers. They do not usually worry about hills so most hills are ignored on a road map. A walker's map is not interested in whether a road is a one-way street, so such details are ignored.

Example 6.10. In medicine, different specialists work with different abstractions of human body. An orthopaedician works with the abstraction of skeletal system, while a gastroenterologist works with digestive system. A physiotherapist abstracts the human body by its muscular system.

We use abstraction in a variety of ways while constructing algorithms — in the specification of problems, representing state